Characteristics and classification of living organisms

20 questions: 10 multiple choice, 10 structured.

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1 · Part 1 Core Characteristics of living organisms [1]

Which statement describes excretion?

A the release of energy from nutrient molecules in cells

B the removal of the waste products of metabolism and substances in excess of requirements

C the removal of undigested food from the body

D the taking in of materials for energy, growth and development

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2 · Part 1 Core Concept and uses of classification systems [1]

Two populations of frogs live on opposite sides of a mountain range. The frogs in the two populations look slightly different.

Which observation shows that the two populations belong to the same species?

A Frogs from both populations eat the same types of insect.

B Frogs from both populations have the same number of toes.

C Frogs from the two populations mate with each other and produce tadpoles.

D Frogs from the two populations mate with each other, and their offspring grow up and produce offspring of their own.

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3 · Part 1 Core Concept and uses of classification systems [1]

The scientific names of four trees are *Pinus sylvestris*, *Malus sylvestris*, *Pinus nigra* and *Fagus sylvatica*.

Which two trees belong to the same genus?

A *Pinus sylvestris* and *Pinus nigra*

B *Pinus sylvestris* and *Malus sylvestris*

C *Malus sylvestris* and *Fagus sylvatica*

D *Pinus nigra* and *Fagus sylvatica*

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4 · Part 1 Core Concept and uses of classification systems [1]

The dichotomous key can be used to identify four types of fish, P, Q, R and S.

step	feature	result
1 (a)	has barbels (whiskers) around the mouth	go to 2
1 (b)	has no barbels around the mouth	go to 3
2 (a)	tail fin is forked	P
2 (b)	tail fin is rounded	Q
3 (a)	body has stripes	R
3 (b)	body has spots	S

One fish has no barbels around its mouth, a forked tail fin and a striped body.

Which fish is it?

A P

B Q

C R

D S

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5 · Part 1 Extended Concept and uses of classification systems [1]

Scientists compared the base sequence of the same gene in species W and in four other species. The table shows the number of bases that differ between species W and each of the other species.

species	number of bases that differ from species W
V	12
X	31
Y	4
Z	18

Which species shares the most recent common ancestor with species W?

A Y

B Z

C X

D V

6 · Part 1 Extended Concept and uses of classification systems [1]

Sharks and dolphins both live in the sea. They have a similar streamlined body shape and both have fins. Sharks are classified as fish and dolphins are classified as mammals.

Which statement explains why sharks and dolphins are placed in different groups?

A Any two animals that cannot interbreed are always placed in different vertebrate groups.

B Classification is based only on the habitat in which an animal lives.

C The DNA of sharks and dolphins is made of different types of bases.

D Classification aims to reflect evolutionary relationships, and sharks and dolphins share only a distant ancestor.

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7 · Part 1 Core Features of organisms [1]

An arthropod has a body divided into two sections, four pairs of jointed legs and no antennae.

To which group does it belong?

A crustaceans

B arachnids

C insects

D myriapods

8 · Part 1 Core Features of organisms [1]

One vertebrate has smooth, moist skin with no scales. It lays eggs without shells in water. Its young live in water and have gills.

To which group does this vertebrate belong?

A fish

B reptiles

C amphibians

D mammals

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9 · Part 1 Extended Features of organisms [1]

Which row could describe the cells of a fungus?

	nucleus	**cell wall**	**chlorophyll**
A	absent	present	absent
B	present	absent	absent
C	present	present	present
D	present	present	absent

10 · Part 1 Extended Features of organisms [1]

Which two structures are features of a virus?

A cell membrane and cytoplasm

B genetic material and a protein coat

C cell wall and genetic material

D nucleus and a protein coat

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11 · Part 1 Core Characteristics of living organisms [5]

One student hangs a small salt crystal in a concentrated salt solution. After one week the crystal is larger. The student also grows a bean seedling, which gets taller each day.

(a) State the name of the characteristic of living organisms described by each statement.

(i) the ability to detect and respond to changes in the internal or external environment [1]

(ii) the chemical reactions in cells that break down nutrient molecules and release energy for metabolism [1]

(b) The student says that the salt crystal is alive because it grows.

Suggest two characteristics of living organisms, other than growth, that the bean seedling shows but the salt crystal does not show. [2]

(c) Growth is a permanent increase in size and dry mass.

Suggest why the dry mass of the seedling, rather than its fresh mass, is measured to find out whether it has grown. [1]

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12 · Part 1 Core Concept and uses of classification systems [6]

The table gives some features of four types of fruit, E, F, G and H.

fruit	fleshy	number of seeds	surface of skin
E	yes	one	hairy
F	yes	many	smooth
G	no	one	smooth
H	yes	one	smooth

(a) Construct a dichotomous key to identify fruits E, F, G and H. Use only the features in the table. [4]

(b) Fruits of the same type can be green, yellow or red.

Suggest why the colour of a fruit would not be a suitable feature to use in this key. [2]

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13 · Part 1 Core Concept and uses of classification systems [6]

The table shows the common names and the scientific names of four mammals.

common name	scientific name
grey wolf	*Canis lupus*
coyote	*Canis latrans*
red fox	*Vulpes vulpes*
Arctic fox	*Vulpes lagopus*

(a) State the genus of the coyote. [1]

(b) Identify the mammal in the table that is most closely related to the red fox. Give a reason for your answer, using information from the table. [2]

(c) Describe what is meant by a species. [2]

(d) The red fox has many different common names in different countries and languages.

Suggest why it is useful for scientists to use the scientific name *Vulpes vulpes*. [1]

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14 · Part 1 Extended Concept and uses of classification systems [6]

Scientists studied four species of beetle, K, L, M and N. They compared the base sequence of a short section of the same gene in each species.

The table shows the base sequences. The bases are written in groups of five to make them easier to compare.

species	base sequence
K	ATGCC TAGGA TCCTA AGCTT
L	ATGCC CAGGA TCCTG AGCTT
M	ATCGC TTGGA GCCAA AACTA
N	ATGAC TAGGT TCCTA AGGTC

The number of bases that differ between species K and species L is 2. The number of bases that differ between species K and species N is 4.

(a) Determine the number of bases that differ between species K and species M. [1]

(b) Identify the species that is most closely related to species K. Explain your answer. [3]

(c) Explain why comparing base sequences in DNA helps scientists to produce classification systems that reflect evolutionary relationships. [2]

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15 · Part 1 Core Concept and uses of classification systems [5]

The dichotomous key can be used to identify the leaves of five types of tree, V, W, X, Y and Z.

step	feature	result
1 (a)	leaf is divided into separate leaflets	go to 2
1 (b)	leaf is not divided into leaflets	go to 3
2 (a)	leaflets all spread out from one point, like fingers on a hand	V
2 (b)	leaflets are arranged in two rows along a central stalk	W
3 (a)	edge of leaf is smooth	X
3 (b)	edge of leaf is toothed	go to 4
4 (a)	leaf is more than twice as long as it is wide	Y
4 (b)	leaf is less than twice as long as it is wide	Z

(a) Use the key to identify each of these leaves.

leaf 1: not divided into leaflets, toothed edge, 8 cm long and 5 cm wide

leaf 2: divided into leaflets arranged in two rows along a central stalk

leaf 3: not divided into leaflets, smooth edge [3]

(b) Suggest why step 4 compares the length of the leaf with its width, instead of using the actual length of the leaf. [1]

(c) State the basis on which scientists classify organisms into groups. [1]

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16 · Part 1 Core Features of organisms [4]

(a) State two features used to place an organism in the animal kingdom. [2]

(b) The Venus flytrap is a plant. Its leaves trap insects and then digest them.

One student says that the Venus flytrap should be classified as an animal because it feeds on other organisms.

Explain why the Venus flytrap is placed in the plant kingdom. [2]

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17 · Part 1 Core Features of organisms [6]

The table describes four arthropods, J, K, L and M.

arthropod	description
J	body in two sections; four pairs of legs; no antennae
K	body in three sections (head, thorax and abdomen); three pairs of legs; one pair of antennae; two pairs of wings
L	body made of many similar segments with legs on every segment; one pair of antennae
M	two pairs of antennae; five pairs of legs; hard exoskeleton covering the body

(a) State the group of arthropods to which each of J, K, L and M belongs. [4]

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(b) One student says that any arthropod with antennae must be an insect.

Explain why the student is wrong. [2]

18 · Part 1 Core Features of organisms [6]

The table describes three vertebrates, F, G and H.

vertebrate	description
F	dry skin covered in scales; lays eggs with soft, leathery shells on land
G	body covered in feathers; has a beak; lays eggs with hard shells
H	skin covered in scales; has fins; breathes using gills

(a) State the vertebrate group to which each of F, G and H belongs. [3]

(b) Bats have wings and can fly.

Explain why bats are classified as mammals and not as birds. [2]

(c) Penguins cannot fly. They use their flippers to swim.

State one feature that a scientist could use to classify a penguin as a bird. [1]

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19 · Part 1 Extended Features of organisms [6]

The table gives some features of four organisms, P, Q, R and S.

organism	number of cells	nucleus	cell wall	chlorophyll	other features
P	one	absent	present	absent	circular DNA and plasmids
Q	many, forming thread-like hyphae	present	present, made of chitin	absent	feeds on dead material
R	one	present	absent	present	lives in pond water
S	many	present	present, made of cellulose	present	has roots and leaves

(a) State the kingdom to which each of P, Q, R and S belongs. [4]

(b) Organism R contains chlorophyll.

Explain why organism R is not placed in the plant kingdom. [2]

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20 · Part 1 Extended Features of organisms [6]

Ferns and flowering plants are two groups within the plant kingdom.

(a) State two features that are used to place a plant in the fern group rather than in the flowering plant group. [2]

(b) The table gives some features of three flowering plants, T, U and X.

flowering plant	leaf veins	number of petals in each flower
T	parallel	6
U	branching network	5
X	branching network	4

State whether each of plants T, U and X is a monocotyledon or a dicotyledon. [3]

(c) Predict the number of cotyledons in a seed of plant T. [1]

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